ECO 344 - Applied Econometrics I
Chapter 8

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Outline

1. Introduction
2. Heteroskedasticity Robust Inference
3. Testing for Heteroskedasticity
4. Weighted Least Squares Estimation
5. The Linear Probability Model: Revisited
6. Summary
7. Class Activity

Introduction

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- For example, in a regression of savings on income, richer households have more room to save a lot or very little, so the spread of savings around the regression line grows with income.
- In cross-sectional data, heteroskedasticity is more the rule than the exception.

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- So, heteroskedasticity is a problem for inference and efficiency. It is not a problem for the estimates themselves.

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5. The confidence intervals based on the usual standard errors are invalid.
6. OLS is no longer efficient (no longer BLUE). There could be another method to estimate the regression coefficients that yields a smaller variance.

Heteroskedasticity Robust Inference

Variance of the OLS Estimator

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$$\text{Var}(\hat{\beta}_j) = \frac{\sum_{i=1}^{n}\hat{r}_{ij}^2\sigma_i^2}{(SSR_j)^2}$$

where $\hat{r}_{ij}$ are the residuals from regressing x_j on all the other explanatory variables and SSR_j is their sum of squares.

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- With a reasonably large sample, robust standard errors are harmless and are the common practice in applied work.

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- Confidence intervals also work as before, $\hat{\beta}_j \pm c \times se_{robust}(\hat{\beta}_j)$.

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- Recall the Chow test from Chapter 7. It is based on the SSR form of the F-statistic, so it is not valid under heteroskedasticity.
- Under heteroskedasticity, test the difference between groups by estimating the model with the group dummies and interactions and conducting a robust Wald test (or robust LM test) on them.

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  4. Regress 1 on $\hat{r}_1\tilde{\mu}, \hat{r}_2\tilde{\mu}, \ldots, \hat{r}_q\tilde{\mu}$ **without an intercept** and get SSR_1.

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  6. Reject H_0 if $n - SSR_1 > \chi^2_{q,\alpha}$ (the critical value), otherwise we are unable to reject H_0.

Example: Heteroskedasticity Robust LM-Test

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- Let's conduct the heteroskedasticity robust LM test.

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The Breusch-Pagan Test for Heteroskedasticity

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- So squares and interactions of the explanatory variables should also be included while testing for heteroskedasticity.

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- Therefore, the model to test becomes:

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- To counter that problem, we can use the special case of the White test, which uses $\hat{y}$ and $\hat{y}^2$ only. The fitted value $\hat{y}$ is a function of all the x's, so $\hat{y}^2$ stands in for all squares and interactions.

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$$H_0 : \delta_1 = \delta_2 = 0, \qquad H_1 : \text{at least one of } \delta_1, \delta_2 \neq 0$$

3. Calculate the F-statistic,

$$F_{calculated} = \frac{R^2_{\hat{\mu}^2}/2}{(1 - R^2_{\hat{\mu}^2})/(n-3)} \sim F_{2,n-3}$$

4. If $F_{calculated} > F_{2,n-3}$ (critical value) then reject H_0 and conclude heteroskedasticity. Otherwise there is no evidence of heteroskedasticity.

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- So, before testing for heteroskedasticity, make sure the model includes the appropriate squared and interaction terms of the explanatory variables and the right transformations.
- A rejection is a sign that something is wrong. It does not tell us by itself what to do.

Book Examples 8.4 and 8.5: Housing Prices

- Example 8.4 estimates a house price equation (hprice1 data) in levels.

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- Example 8.5 applies the special case of the White test to this log model. Is the evidence of heteroskedasticity still there?
- In the level model we reject homoskedasticity, while in the log model we do not. Taking logs is a simple and often effective way to reduce heteroskedasticity.

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- The rest of the chapter covers the second option.

Weighted Least Squares Estimation

Efficiency of the Estimator

- Recall, under M.L.R - I through M.L.R - V, the OLS estimators are Best Linear Unbiased Estimators (Gauss-Markov). With M.L.R - VI added, we also get exact t- and F-distributions.

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- Under heteroskedasticity, M.L.R - V is violated. Robust standard errors fix the inference, but OLS is no longer efficient.
- Therefore, we should look for another method that has a lower variance.

When the Heteroskedasticity Function is Known

- Suppose the variance of the error depends on a known function of the explanatory variables.

$$\text{Var}(\mu|x) = \sigma^2 h(x)$$

where $h(x) > 0$ is some function of the explanatory variables and σ^2 is an unknown parameter we can estimate from the sample.

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- Divide both sides of the regression model by $\sqrt{h(x)}$. The model becomes

$$\frac{y}{\sqrt{h(x)}} = \beta_0 \frac{1}{\sqrt{h(x)}} + \beta_1 \frac{x_1}{\sqrt{h(x)}} + \cdots + \beta_k \frac{x_k}{\sqrt{h(x)}} + \frac{\mu}{\sqrt{h(x)}}$$

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- and the new error has a constant variance,

$$\text{Var}\left(\frac{\mu}{\sqrt{h(x)}}\middle| x\right) = \frac{\sigma^2 h(x)}{h(x)} = \sigma^2$$

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- In software, give the weight $\frac{1}{h(x)}$. In Stata, add "[aweight = 1/h]" to the regression command. In R, use "weights = 1/h" in lm().

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- Example 8.6 estimates net total financial wealth for single-person households (401ksubs data).

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- $h(x) = inc$, so the weight is $\frac{1}{inc}$. Households with higher income get less weight.
- Compare the OLS and WLS estimates and standard errors. Are the coefficients similar? Are the standard errors different?

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- This method is known as Feasible GLS (FGLS).

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5. Estimate the original model by WLS using the weights $\frac{1}{\hat{h}}$.

   (Equivalently, divide every variable, including the constant, by $\sqrt{\hat{h}}$ and run OLS.)

Another Way to Estimate the Heteroskedasticity Function

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- Now, we can estimate the FGLS by using $\frac{1}{\hat{h}}$ as the weights.
- Similarly, there are other methods to estimate $\hat{h}$.

Properties of FGLS

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- The price is that the results now depend on getting the variance function right.

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- Example 8.7 estimates the demand for cigarettes (smoke data).

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- Compare the OLS and FGLS estimates and standard errors. Which coefficients change, and does any conclusion change?

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- Prediction under heteroskedasticity needs $\text{Var}(\mu|x_0) = \sigma^2 h(x_0)$ at the point of interest x_0, so the prediction interval is wider for points with a high error variance.

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- Therefore, we should always use robust standard errors (and robust tests) when estimating the LPM.
- We do not need to test for heteroskedasticity in the LPM. We already know it is there.

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- Book Example 8.8 (PC ownership, gpa1 data) estimates an LPM and compares OLS with robust standard errors to WLS.

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- Report OLS with robust standard errors alongside any WLS/FGLS estimates.
- The LPM is always heteroskedastic. Use robust standard errors, and WLS only when all fitted values lie strictly between 0 and 1.

Class Activity